Final Project Proposal

Statistical Programming Paradigms and Workflows (BSPH 140.777)

Team Git Committee

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GitHub Repository: https://github.com/stephaniemiless/paradigms_git_committee/tree/main

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Research Question and Audience

Research Question: Is there an association between air quality and asthma flare-ups (described as Emergency Department Visits/Hospitalizations for asthma) in the United States at the State level? Which states have stronger associations?

Why does this matter? Climate change exacerbates the health impacts of poor air quality. The WHO estimates that air pollution is responsible for 7 million premature deaths annually, making it one of the largest global risk factors for death, trailing only high blood pressure, smoking, and diabetes.[1] Climate change drives frequency and intensity of wildfires, increased tropospheric ozone, and stagnant air particularly in urban areas. Particulate matter and smog cause acute respiratory damage and increase asthma prevalence.[2] Through this dashboard, we hope to evaluate how air quality and asthma rates vary geographically, which could inform at-risk populations, national economic impact, and potential policy interventions.

Intended Audience: As we are limiting our geographical region to the US at the state level, the intended audience would be people living in the US. Ideally, users can use it every day to check recent air quality statuses and predict those for the next few days as well.

Related Work

There are many different works involved with working with the air quality index and asthma or respiratory flare ups (which we refer to as emergency department visits and hospitalization). The Air Quality Index (AQI) has been used as an indicator of air quality and provides behavioral recommendations to reduce exposure and therefore harm. The University of Pittsburgh has conducted a test for association between the AQI and childhood asthma flare ups resulting in emergency department visits and hospitalizations. They used data from the Environmental Protective Agency (EPA) which was obtained for Allegheny

[1] Council on Foreign Relations. How climate change affects our health. Accessed November 13, 2026. https://www.cfr.org/education/learn/reading/how-climate-change-affects-our-health

[2] US Environmental Protection Agency. Air quality and climate change research. Accessed November 13, 2026. https://www.epa.gov/air-research/air-quality-and-climate-change-research

County, Pennsylvania. They found that the AQI was associated with asthma flare ups among children in Allegheny County, primarily affecting Black and younger children.[3]

Another group assessed associations between emergency room visits for asthma and chronic obstructive pulmonary disease (COPD) with the zip code level of exposure to criteria air pollutants in New York State. This study utilizes the EPA's National Emission Inventory database and ER visits to assess the correlation between both the chronic (asthma and COPD) and acute (asthma attacks) events that lead to ER visits. This study was limited to using only point source (fixed, stationary industrial and commercial facilities that release emissions) exposure data to be able to use the ER data at a zip code level.[4] This study also adjusted the data on the poverty rates by zip code, which is an interesting factor we hope to involve in our project.

A tool that is currently being used is called the BenMAP-CE which helps to estimate the health and economic effects of pollution. It predicts the number of and economic value of health impacts that are a result of changes in air quality. The economic values due to air quality change are calculated using two different metrics. One is called "Cost of Illness" which takes into account the expenses that someone must bear for air pollution related hospital admissions and ER visits which includes the value of medical expenses as well as lost wages. On the other hand, the "Willingness to Pay" metric accounts for the direct costs as well as pain and suffering, loss of satisfaction, and leisure time.[5]

Throughout the existing literature, a consistent finding emerges—poor air quality has a significant association with increased emergency department visits and hospitalizations for asthma and other respiratory conditions. Short-term exposure to air pollutants especially contribute to these adverse health outcomes, resulting in substantial healthcare utilization and socioeconomic burden.[6]

The CDC Heat & Health[7] dashboard shows a heatmap of daily heat-related illness for the United States. This is an example dashboard that we would want our own to somewhat resemble. Users are allowed to select the current or a past date to customize the heatmap display. There is an option to enter ZIP codes or county names to take a closer look at the data for the specified region. For example, when 21205 (East Baltimore) is entered, the dashboard outputs the day's heat risk, heat forecast for the next week, a graph comparing daily maximum temperature to historical averages, a heatmap with different heat health index options, and the region's demographics.

Dr. Datta at JHU has made a Baltimore Air Quality Dashboard[8], which displays the current PM2.5 level, the weather, the PM2.5 level variations by day and hour, and a static as well as a dynamic heatmap of the PM2.5 level across the city. Historical data is contained in a separate tab with options to view them in the form of an area map, a city map, or informative statistical graphs.

After careful review of such dashboards, we would want ours to display similar customizations and

[3]Rosser F, Han YY, Rothenberger SD, Forno E, Mair C, Celedón JC. Air quality index and emergency department visits and hospitalizations for childhood asthma. Ann Am Thorac Soc. 2022;19(7):1139-1148. doi:10.1513/AnnalsATS.202105-539OC

[4]Lebbie TS, Jones LE, Madani NA, Carpenter DO. Associations between emergency room visits for respiratory diseases and exposure to zip code-level criteria air pollutants in New York State. Atmosphere. 2026;17(3):322. doi:10.3390/atmos17030322

[5]US Environmental Protection Agency. How BenMAP-CE estimates the health and economic effects of air pollution. Published May 20, 2026. Accessed November 13, 2026. https://www.epa.gov/benmap/how-benmap-ce-estimates-health-and-economic-effects-air-pollution

[6]Zheng XY, Ding H, Jiang LN, et al. Association between air pollutants and asthma emergency room visits and hospital admissions in time series studies: a systematic review and meta-analysis. PLoS One. 2015;10(9):e0138146. doi:10.1371/journal.pone.0138146

[7]Centers for Disease Control and Prevention. Heat & Health Tracker. Accessed November 13, 2026. https://ephtracking.cdc.gov/Applications/heatTracker/

[8]Johns Hopkins University. Baltimore Air Quality Dashboard. Accessed November 13, 2026. https://web.idies.jhu.edu/airquality/dashboard/realtime

information. It is a goal for our final product to be user-friendly and adaptable to the location the user resides in. However, our group's final product would incorporate data such as region-specific asthma flare-ups and socioeconomic statuses to add dimension to our analysis and exploration.

Although existing research has demonstrated associations between outdoor air pollution and asthma related health outcomes, many studies have focused on individual counties, cities, or states rather than examining these relationships across the United States. A nationwide analysis could provide a broader understanding of how these associations vary geographically and whether certain states experience disproportionately greater health impacts. Additionally, differences in socioeconomic conditions, population demographics, and access to healthcare may contribute to variations in asthma-related healthcare utilization across regions. By combining air quality measurements with state-specific health and socioeconomic data, our project aims to build upon existing research by examining both the overall association between outdoor air pollution and asthma-related outcomes and the geographic variation in these relationships.

Data and Access

We plan on getting our data from the Centers for Disease Control and Prevention's (CDC) National Environmental Public Health Tracking Network API. This API contains national, state, and county-level data on Asthma prevalence for adults and children, Emergency Department Visits for asthma, and hospitalizations for asthma. This API is free and accessible to use, which allows us to retrieve the JSON data and parse it into an R data set that we will inspect for missing values.

We additionally plan to request the Air quality data from the CDC National Environmental Public Health Tracking Network API since this API contains county-level data for pollutants such as ozone levels and PM2.5 concentration that would serve useful. From this API, we will retrieve the JSON data and parse it into an R data set that we will further clean by removing missing values. If the data for a county or state of interest is not available in the CDC API, we plan on retrieving the data from the USA Environmental Protection Agency's API on air quality. This API is free to retrieve data after creating an account, and we will retrieve the JSON data and import it into R. From there, we will account for missing values.

To incorporate social determinants of health into our analysis, we also plan to retrieve demographic, socioeconomic, and housing data from the U.S. Census Bureau's API, primarily using the American Community Survey (ACS) 5-Year Estimates. We will request a U.S. Census Data API Key. This dataset provides county-level information on population characteristics, median household income, poverty rates, educational attainment, employment, health insurance coverage, racial and ethnic composition, and housing conditions. These indicators will allow us to investigate how socioeconomic characteristics may relate to differences in environmental exposures and asthma-related health outcomes across communities.

API requests will specify the relevant ACS variables and geographic level, returning JSON data that can be converted into R data frames. We will select relevant demographic and economic indicators, standardize variable names and data types, and reshape the data into a wide format with one observation per county and separate columns for each indicator. We will retain available estimates and margins of error to account for the uncertainty associated with survey-based estimates. We will create a unified county-level dataset containing air quality measurements, asthma prevalence and healthcare utilization, and socioeconomic characteristics. We will use this combined dataset to examine potential relationships between social determinants of health, environmental exposures, and asthma burden. For example, we could investigate whether counties with higher poverty rates, lower household incomes, or lower health insurance coverage experience different asthma-related health outcomes at comparable levels of air pollution. These analyses will help us identify communities that may face overlapping environmental and socioeconomic vulnerabilities and provide additional context for interpreting the geographic distribution

of asthma burden.

API Links:

1. USA EPA API: https://docs.airnowapi.org/
2. CDC ER Data: https://ephtracking.cdc.gov/DataExplorer/
3. Demographic/Economic API: https://www.census.gov/data/developers/data-sets.html

Planned Work

1. General EDA on EPA API air quality and CDC ER datasets
2. Use machine learning to make predictions about asthma prevalence
3. Develop Shiny dashboard

After importing the data into R, we plan on accounting for missing values within counties and states to identify the gaps within our analysis. After cleaning the data for missing values, we plan on creating figures such as histograms and scatterplots that we believe would be pertinent to include in the R Shiny dashboard. We plan on finding descriptive statistics, such as the mean pollutant levels in states by looking at county data and using this to present the states with the highest mean air pollutant levels. We also plan on creating figures displaying the states with the highest prevalence of asthma stratified by age group. For the figures portraying the associations, we plan to utilize either scatterplots of the distributions or histograms depending on what the data looks like. On our interactive dashboard, we plan on having options where users can decide which figures they would like to look at such as specifying a metric or state they are interested in.

Programming Paradigms

We aren't 100% knowledgeable of the exact tasks we will be carrying out for this project, but we do believe that our programs will incorporate the following paradigms:

- **Functional programming**: Through data cleaning and other analyses, there is bound to be repeated code that could be refactored through resuable operations and mapping operations. Packages like `purrr` and `rvest` could be utilized here.
- **Declarative programming**: Through data cleaning, we will certainly be utilizing tools such as `dplyr` to carry out the work that we want to do. For example, we will probability utilize `tidyverse` heavily for the data cleaning and EDA portions of the project.
- **Machine Learning**: One of the more complex portions of this project includes making predictions based on the three data sets we have. We aren't entirely sure on the specific models we want to use to carry this out, but we do know that machine learning will take a big part in the final product. For example, geographic/AQI/demographic info can shape predictions about asthma prevalence.

Software

- Programming Language: R
- Packages: Shiny, ggplot, dplyr, tidyverse, machine learning packages (ie. randomForest, xgboost, caret)

Product

We will develop an interactive R Shiny dashboard that allows users to explore the geographic distribution of air pollution and asthma burden across the United States. The dashboard will integrate environmental data on air pollutants, including PM2.5 and ground-level ozone, with public health indicators such as asthma prevalence among adults and children, asthma-related emergency department visits, and hospitalizations at the state and county levels, where available. Additionally, we will incorporate demographic, socioeconomic, and housing information, including income, poverty rates, educational attainment, health insurance coverage, and population characteristics, to examine how social determinants of health may relate to environmental exposures and asthma outcomes. Users will be able to select geographic locations, including individual states and counties, and stratify results by demographic characteristics to explore differences in environmental exposures and asthma-related health outcomes across populations. The dashboard will offer customizable data visualizations and potentially interactive exploratory data analysis (EDA), allowing users to select variables, compare indicators, and investigate patterns within the integrated datasets. We also plan to incorporate a machine learning component using interpretable predictive models to explore whether asthma prevalence and related health outcomes can be predicted using environmental, geographic, and demographic characteristics. For example, users could examine how county-level air pollution concentrations, population characteristics, and socioeconomic indicators relate to predicted asthma prevalence. The dashboard will be designed for public health researchers, policymakers, environmental health professionals, and the general public. We plan to publish the application on the Shiny hosting server, making it publicly accessible through a web browser without requiring users to install R or have programming experience.

Timeline and Feasibility

Tentative Deadlines:

1. ***Tuesday, Nov 17***: Initial meeting with Prof. Hicks to discuss project proposal
2. ***Friday, Nov 20***: Identify, access, and download necessary data across asthma, air quality, population demographics, and economic impacts.
3. ***Monday, Nov 30***: All data sets cleaned by this point
4. ***Thursday, Dec 10***: Drafted product done
5. ***Thursday, Dec 17***: In-class team presentation

For the 5 remaining weeks of the term, we plan on setting goals for the tasks that we would like to get done. In the first week, we aim to import the JSON data into R and come up with a final analysis dataset, which we will use to create the figures that will be available on our dashboard. During the second week, we plan to finalize the relevant associations, such as the association between Ozone levels and asthma-related emergency room hospitalizations. During the third and fourth weeks, we plan to finalize the R Shiny dashboard in preparation for the final project presentation and to ensure that the results are accessible.

Potential Challenges

Potential challenges to tackling our research question are that for some metrics like asthma hospitalizations or emergency department visits, there are some states with no data for certain time periods. This could make it difficult to quantify flare-ups for the United States as a whole, but we would work to quantify flare-ups for the states that we do have and provide prevalence statistics for the other states.

Missing data for some states also makes it difficult to implement machine learning models on the national scale, but for this model we plan on implementing it using available data from certain states.

Manageable initial version of the product:

An interactive R Shiny dashboard where users can see relevant figures on the presence of air pollutants, prevalence of asthma by state and age status, and figures on the association between air pollutant levels and asthma-related emergency room visits and hospitalizations. A baseline analysis that uses general machine learning topics such as clustering and regression to discover underlying patterns and make predictions about the data.

Team Responsibilities

Each team member will work collaboratively within our shared repository on their various responsibilities. Each individual will work on all parts of the project, but we decided to assign task leaders to serve as a point of contact for designated tasks.

- **Steph**: Point person for the Shiny dashboard portion of the project, lead for machine learning aspect
- **Fabiana**: Clean the asthma data from the CDC API. Work on a significant portion of the figures that will be put on the dashboard (act as the EDA point person).
- **Eva**: Point person for cleaning the economic/census data set, work on machine learning aspect with Allion and Steph.
- **Allison**: Lead for machine learning aspect, work on EDA with Fabiana
- **Ania**: Slides, point person for cleaning the air quality data set

The team will communicate daily through a groupchat and everyone will stay on top of updating the repository every time they make a substantial change.